AWS VPC PEERING
STUDENT PROJECT REPORT

Secure private connectivity between two Amazon Virtual Private Clouds

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Batch: AWS Internship Batch1

AWS Region: Asia Pacific (Mumbai) — ap-south-1

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1. Project Overview

This project demonstrates private communication between two Amazon VPCs using VPC Peering. VPC A contains a public bastion and a private application host; VPC B contains a private application host. Route tables and security groups control traffic. Region: Mumbai (ap-south-1).

2. Objectives

• Create two non-overlapping VPCs and configure subnets.

• Establish VPC peering and configure forward and return routes.

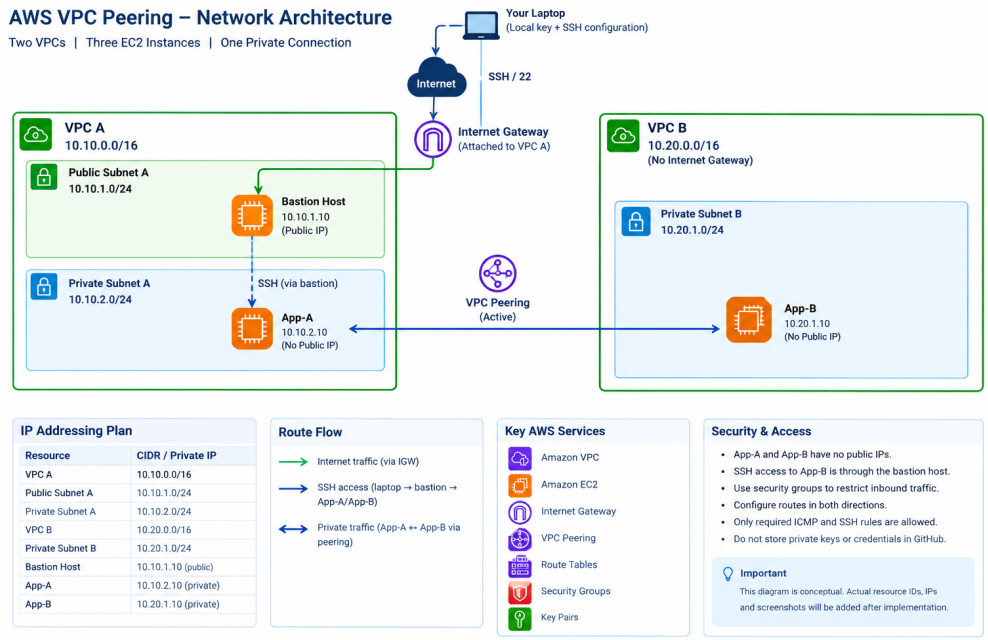
• Restrict inbound traffic with security groups.

• Verify bidirectional ICMP and private SSH access through the bastion.

• Document actual results and clean up lab resources.

3. Architecture and Addressing

VPC A (10.10.0.0/16): public subnet 10.10.1.0/24 and private subnet 10.10.2.0/24. VPC B (10.20.0.0/16): private subnet 10.20.1.0/24.



Resource	Name / ID	CIDR / Address
VPC A	ewb-vpc-a / vpc-069ab77a8167915a5	10.10.0.0/16
VPC B	ewb-vpc-b / vpc-0cac7153b279dc807	10.20.0.0/16
Public subnet A	ewb-a-public / subnet-0449d69d8404dddbf	10.10.1.0/24
Private subnet A	ewb-a-private / subnet-01c558669ecd54a5d	10.10.2.0/24
Private subnet B	ewb-b-private / ID to confirm	10.20.1.0/24
Internet gateway	ewb-igw-a / igw-0508230ad22b0724f	Attached to VPC A
Bastion	ewb-bastion	10.10.1.179; public 13.232.37.153
App-A	ewb-app-a	10.10.2.165; no public IPv4
App-B	ewb-app-b	10.20.1.42; no public IPv4
Peering	ewb-a-b-peer / ID to confirm	Status reported Active

4. Route Tables

Route table	Association	Routes
ewb-rt-a-public	ewb-a-public	10.10.0.0/16 → local; 0.0.0.0/0 → IGW; 10.20.0.0/16 → peering
ewb-rt-a-private	ewb-a-private	10.10.0.0/16 → local; 10.20.0.0/16 → peering
ewb-rt-b-private (rtb-0e423296f55a2b12f)	ewb-b-private	10.20.0.0/16 → local; 10.10.0.0/16 → peering

Forward path: App-A sends to 10.20.1.42. VPC A's private route table sends 10.20.0.0/16 to the peering connection; VPC B's local route delivers traffic to App-B.

Return path: App-B's response to 10.10.2.165 follows VPC B's 10.10.0.0/16 peering route, then VPC A's local route delivers it to App-A.

5. Security Groups

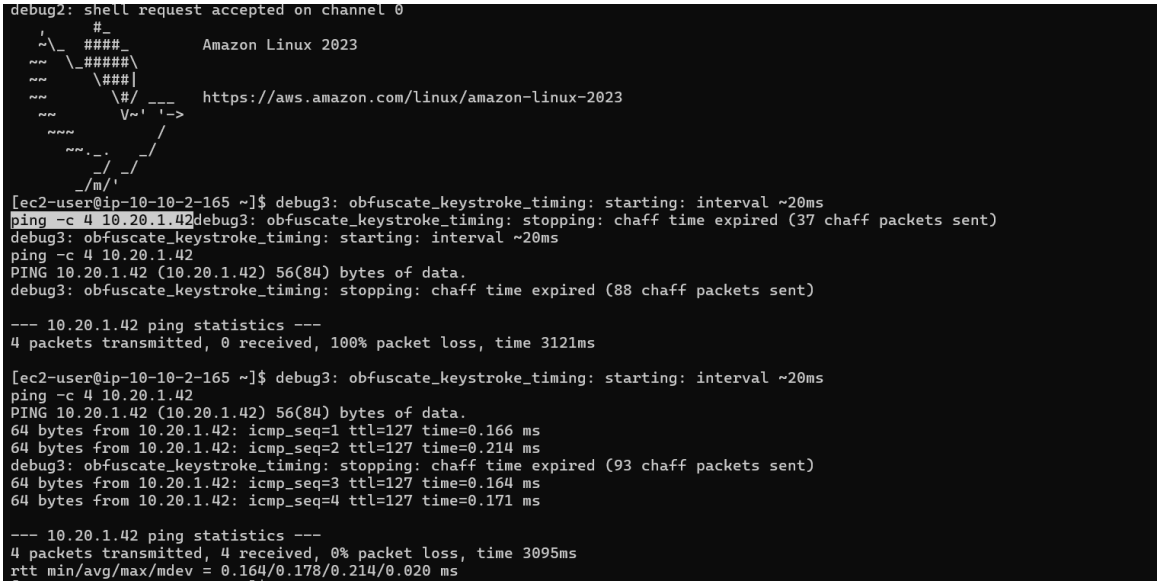
Security group	Inbound rules / intended restriction
ewb-sg-bastion	SSH TCP/22 from user's public IP /32 only.
ewb-sg-app-a	SSH TCP/22 from 10.10.1.179/32; ICMP Echo Request from 10.20.1.42/32.
ewb-sg-app-b	SSH TCP/22 from 10.10.1.179/32; ICMP Echo Request from 10.10.2.165/32.

Verify the final inbound rules in AWS Console before submission.

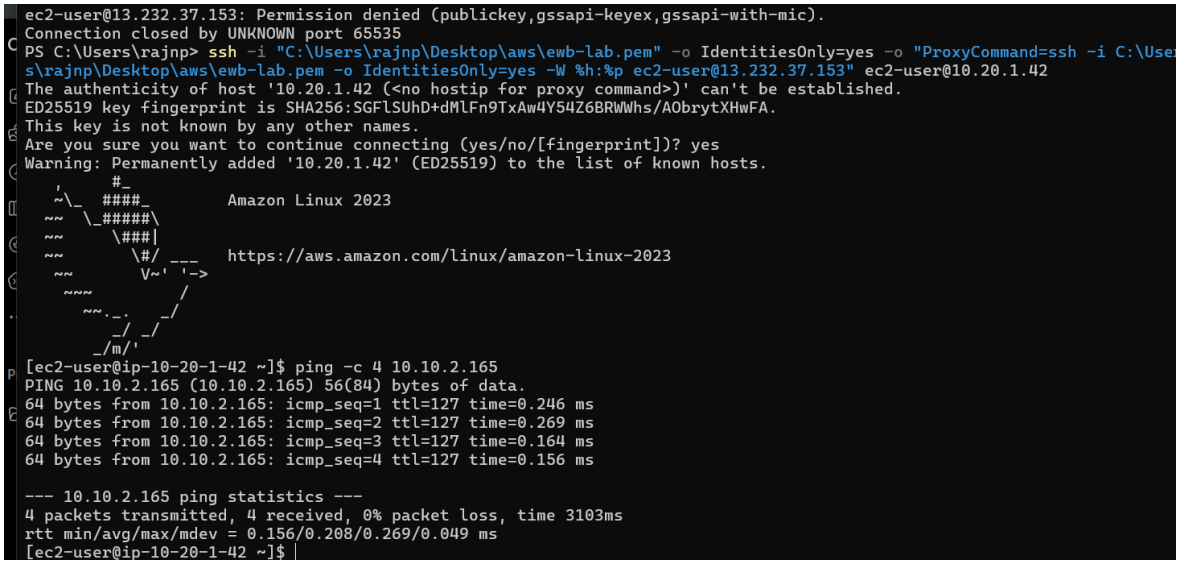
6. Connectivity Test Results

Test	Result / evidence status
App-A → App-B ping	Verified: 4 received, 0% packet loss. Screenshot collected.
App-B → App-A ping	Verified: 4 received, 0% packet loss. Screenshot collected.
SSH to App-A via bastion	Previously confirmed successful by user; attach terminal evidence if available.
SSH to App-B via bastion	Previously confirmed successful by user; attach terminal evidence if available.
Baseline failure before peering	Not captured before peering was configured.
Controlled ICMP failure / recovery	Recovery previously reported; failure screenshot skipped.
Private-host public IPv4 check	User reported both private hosts have no public IPv4; add console evidence if available.

Forward ping evidence



Reverse ping evidence



7. Access Design and Security

The bastion is the only instance with a public IPv4 address. App-A and App-B are accessed using SSH ProxyCommand. Keep the private key on your computer; never upload it, credentials, or tokens to GitHub.

8. Limitations and Evidence Gaps

• The baseline failure was not captured before peering.

• The controlled ICMP failure screenshot was skipped; no screenshot is claimed.

• Confirm the Private subnet B ID, peering ID, route tables, associations, and security-group rules in AWS Console.

• Add SSH, isolation, and cleanup evidence where available.

9. Cleanup and Cost Control

After saving evidence, terminate ewb-bastion, ewb-app-a, and ewb-app-b. Check for leftover EBS volumes, snapshots, and Elastic IPs. Delete the peering connection and peer routes, security groups after network interfaces disappear, subnets, custom route tables, internet gateway, and both lab VPCs. Verify no lab resources remain and check Billing; usage reporting may lag. Cleanup evidence has not yet been added.

10. Learning Outcomes

• VPC peering does not automatically configure routes.

• Forward and return routes are both required.

• Security groups must allow intended inbound traffic.

• A bastion provides controlled SSH access to private hosts.

• Evidence should reflect actual execution.

11. Final Submission Checklist

■ Confirm missing resource IDs.

■ Verify route tables, associations, and security groups.

■ Add SSH/isolation evidence where available.

■ Add cleanup evidence after deletion.

■ Do not include private keys, passwords, tokens, or access keys.